

Another measure of importance in a multivariate problem is the partial correlation. Partial correlation coefficient provides a measure of the relationship between the dependent variable and other variable, with the effect of the rest of the variables eliminated.

If there are three variables X_1, X_2 and X_3 , there will be three coefficients of partial correlation, each studying the relationship between two variables when the third is held constant. If we denote by $r_{12.3}$ that is, the coefficient of partial correlation X_1 and X_2 keeping X_3 constant, it is calculated as:

$$r_{12.3} = \frac{r_{12} - r_{13}r_{23}}{\left(\sqrt{1 - r_{13}^2}\right)\left(\sqrt{1 - r_{23}^2}\right)}. \text{ 'Similarly}$$

r_{xy} - correlation coefficient
between variables X, Y .

$$r_{13.2} = \frac{r_{13} - r_{12}r_{23}}{\left(\sqrt{1 - r_{12}^2}\right)\left(\sqrt{1 - r_{23}^2}\right)}, \text{ and}$$

$$r_{23.1} = \frac{r_{23} - r_{12}r_{13}}{\left(\sqrt{1 - r_{12}^2}\right)\left(\sqrt{1 - r_{13}^2}\right)},$$

where r_{12}, r_{23} and r_{13} is the correlation coefficient between respective two variables.

Problem: Suppose on the basis of observations on hens in poultry, the number of eggs laid by hens (X_1) depends on food (X_2), and types of hen (X_3). It is found that $r_{12} = 0.8$, $r_{13} = 0.65$ and $r_{23} = 0.7$ Find the $r_{12.3}$, $r_{13.2}$ and $r_{23.1}$.

Soln:-

$$r_{12.3} = \frac{r_{12} - r_{13} r_{23}}{\sqrt{1 - r_{13}^2} \sqrt{1 - r_{23}^2}} = \frac{0.8 - (0.65)(0.7)}{\sqrt{1 - (0.65)^2} \sqrt{1 - (0.7)^2}} = 0.635$$

$$r_{13.2} = \frac{r_{13} - r_{12} r_{23}}{\sqrt{1 - r_{12}^2} \sqrt{1 - r_{23}^2}} = \frac{0.65 - (0.8)(0.7)}{\sqrt{1 - (0.8)^2} \sqrt{1 - (0.7)^2}} = 0.21$$

$$r_{23.1} = \frac{r_{23} - r_{12} r_{13}}{\sqrt{1 - r_{12}^2} \sqrt{1 - r_{13}^2}} = \frac{0.7 - (0.8)(0.65)}{\sqrt{1 - (0.8)^2} \sqrt{1 - (0.65)^2}} =$$

Multiple correlation

In multiple correlation, we are trying to make estimates of the value of one of the variable based on the values of all the others. The variable whose value we are trying to estimate is called the dependent variable and the other variables on which our estimates are based are known as independent variables.

The coefficient of multiple correlation with three variables X_1, X_2 and X_3 are $R_{1.23}, R_{2.13}$ and $R_{3.21}$.

$R_{1.23}$, is the coefficient of multiple correlation related to X_1 as a dependent variable and X_2, X_3 as two independent variables and it can be expressed in terms of r_{12}, r_{23} and r_{13} as below

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{23}r_{13}}{1 - r_{23}^2}}$$

$$R_{2.13} = \sqrt{\frac{r_{12}^2 + r_{23}^2 - 2r_{12}r_{23}r_{13}}{1 - r_{13}^2}}$$

$$R_{3.12} = \sqrt{\frac{r_{13}^2 + r_{32}^2 - 2r_{12}r_{23}r_{13}}{1 - r_{12}^2}}.$$

Problem: A simple correlation coefficient between yield (X_1), temperature (X_2) and rainfall (X_3) are given by $r_{12} = 0.6$, $r_{13} = 0.5$ and $r_{23} = 0.8$
Calculate $R_{1.23}$, $R_{2.13}$ and $R_{3.21}$

Soln:-

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{23}r_{13}}{1 - r_{23}^2}} = \sqrt{\frac{(0.6)^2 + (0.5)^2 - 2(0.6)(0.8)(0.5)}{1 - (0.8)^2}}$$
$$= 0.60.$$

Calculate $R_{2.13}$, $R_{3.21}$ (Home work).